MODULE 15 · DOMAIN 4 - WORKFLOW INTEGRATION AND SOLUTION DESIGN

Research, Planning and Process Optimisation

Questions before sources, objectives before tasks, and measurement before any fix.

DOMAIN 4 of 7 EXAM WEIGHT 16% MODULE 15 of 33

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Research, plans and process fixes all **start with framing**, not with action.

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Research done well

01
Frame
The decision it serves, and the questions.

02
Scope
Sources that count, timeframe, what is out.

03
Gather
Research, web search, your own documents.

04
Rate
Sources and confidence for each finding.

05
Verify
The claims the decision rests on.

06
Summarise
For the decision, with gaps stated.

ON THE EXAM

Questions before sources. Confidence before conclusions.

Asking for sources and confidence

- **A source for every finding,** and a confidence rating: high, medium or low, with the reason.
- **Separate fact from inference:** what a source says, and what Claude concludes from it.
- **Show disagreement:** where sources conflict, and which is primary.
- **Say what was not found.** A gap stated plainly beats a gap filled with a guess.

> THE TIEBREAKER
>
> A finding without a source and a confidence level is an opinion.

Which research tool for which question?

THE QUESTION	REACH FOR	NOTE
One current fact: a rate, a date, a rule	Web search	New experience: Claude searches when it helps
A multi-source investigation with citations	Research	Takes minutes, uses limits faster
What your own documents say	Upload, a Project, or a connector	Internal evidence the web cannot see
Numbers across a large dataset	Code execution	Calculated, not estimated

THE TIEBREAKER

Match the tool to the question, not the question to the tool you like.

Planning with Claude

01
Objective
Outcome, scope and constraints.

02
Phases
Milestones and what ends each phase.

03
Tasks
Within each phase, with dependencies.

04
Risks
Assumptions, risks and mitigations.

05
Commit
People confirm owners, dates and review points.

ON THE EXAM

Objective and phases before detail. Claude drafts the plan; people commit to it.

Discovery interviews with Claude

- **Before:** Claude drafts an interview guide with open, non-leading questions tied to your objective.
- **After:** label each transcript, then ask for themes with supporting quotes and who said them.
- **Frequency is not importance.** One safety concern can outweigh ten comments about the kettle.
- **Get consent to record,** and remove personal details the analysis does not need.

> THE TIEBREAKER
>
> Themes with quotes and sources, judged by a person, not counted by a machine.

Process optimisation: the sequence

01
Map
Each step as it really runs.

02
Measure
Time at each step, from real data.

03
Find the bottleneck
The step that holds the most time.

04
Options
Claude proposes; people choose.

05
Pilot
Small, against the baseline.

06
Decide
Scale, adjust or stop.

ON THE EXAM

Map and measure before you fix. Pilot before you roll out.

Quirk 14: the slow intake, measured

STEP	MEDIAN TIME	SHARE OF TOTAL
Client reports issue, ticket logged	0.5 days	10%
Ticket triaged: category and priority	2.5 days	50%
Quote approved, where needed	1.0 day	20%
Engineer booked	1.0 day	20%
Total, report to booking	5.0 days	100%

THE SIGNAL

Claude worked out the medians from three months of ticket timestamps, using code execution.

Getting the bottleneck numbers right

- **Agree the definitions first:** when does triage start, and when does it end?
- **Export the timestamps** and have code execution calculate each step, not Claude's head.
- **Use medians as well as averages.** One ticket stuck for a month distorts an average.
- **Sanity-check by hand:** trace a handful of tickets end to end and compare.

> ON THE EXAM
>
> Defined, calculated, sanity-checked: then the bottleneck is evidence, not opinion.

Fixing by opinion versus fixing by measurement

BY OPINION

Hire more engineers

- The loudest complaint wins
- Fixes a step that holds 20%
- Expensive and slow to reverse
- No baseline, so no proof

BY MEASUREMENT

Fix triage first

- The data names the bottleneck
- Targets the step that holds 50%
- Options piloted in one region
- Same measure before and after

THE TIEBREAKER

Optimise the slowest step, not the loudest one.

Piloting against a baseline

- **Record the baseline before changing anything.** Same measure, same definition, before and after.
- **Pilot small:** one region or team, for a fixed period, with a named owner.
- **Measure quality as well as speed:** faster triage with wrong priorities is not an improvement.
- **Decide on evidence:** scale, adjust or stop, and write down why.

> ON THE EXAM
>
> A pilot without a baseline proves nothing. A pilot that only measures speed proves half.

Research, planning and process traps

THE TRAP	WHY IT FAILS
Searching before framing the question	Lots of material, no answer to the real decision
Trusting a confident, uncited finding	Fluency is not evidence
A detailed task list with no objective	Detail that serves nothing in particular
Dates and owners nobody agreed	A wish, not a plan
Fixing the loudest step	Effort lands where the time is not
Rolling out with no pilot or baseline	No proof, and failures hit everyone at once

ON THE EXAM

Every trap acts before it frames, or decides before it measures.

> Halden's operations director must advise the board on whether to trial robotic floor cleaners at large retail sites.
>
> She plans to use Claude to research the question.

What should she do first?

A Write down the decision the research serves, the questions and which sources count

B Start a Research task on cleaning robots straight away

C Ask Claude which robot supplier Halden should choose

D Gather every article on cleaning robots she can find

◆ PAUSE AND DECIDE

What should she do first?

A Write down the decision the research serves, the questions and which sources count

Framing: the research now serves a real decision.

B Start a Research task on cleaning robots straight away

Acts before framing; the search has no target.

C Ask Claude which robot supplier Halden should choose

Jumps to a decision that is not Claude's, and skips the research.

D Gather every article on cleaning robots she can find

Volume without a question to answer.

THE SIGNAL

Research serves a decision: frame the questions before gathering anything.

> Claude's research summary says robotic cleaners "cut labour costs by 40%". The citation is a trade blog, which quotes a study paid for by a robot vendor.
>
> The figure will appear in the board paper.

What is the best way to handle this figure?

A Keep it as it is, because it has a citation

B Delete it and say nothing about labour costs

C Ask Claude to reword it so it sounds less certain

D Trace it to the vendor study, check what was measured, and report it as vendor-sourced with its confidence

◆ PAUSE AND DECIDE

What is the best way to handle this figure?

A Keep it as it is, because it has a citation

A citation to a secondhand source is not verification.

B Delete it and say nothing about labour costs

Loses evidence the board may need; state it honestly instead.

C Ask Claude to reword it so it sounds less certain

Changes the tone, not the evidence.

D Trace it to the vendor study, check what was measured, and report it as vendor-sourced with its confidence

Primary source checked, origin and confidence stated.

THE SIGNAL

A secondhand figure from an interested party: go to the primary source, then label it.

> Halden's operations director is planning the roll-out of a new digital timesheet to all field staff, after a pilot at three sites.
>
> She wants Claude's help to build the plan.

Which two steps belong at the start of planning? Choose two.

A State the objective, scope and constraints, such as budget, go-live date and staff consultation

B Ask Claude to assign named owners and dates to every task

C Agree the phases and milestones before breaking them into tasks

D Ask Claude for a detailed list of two hundred tasks straight away

E Ask Claude to confirm the plan with the regional directors

◆ PAUSE AND DECIDE

Which two steps belong at the start of planning? Choose two.

A State the objective, scope and constraints, such as budget, go-live date and staff consultation

The objective and constraints frame everything after.

B Ask Claude to assign named owners and dates to every task

Owners and dates are committed by people, at the end.

C Agree the phases and milestones before breaking them into tasks

Phases before detail.

D Ask Claude for a detailed list of two hundred tasks straight away

Detail before structure, with nothing to hang it on.

E Ask Claude to confirm the plan with the regional directors

Confirmation is a human conversation, and it comes last.

THE SIGNAL

Objective first, then phases. Detail, owners and sign-off come later, from people.

Quirk 14: Halden's reactive maintenance tickets take five days from client report to engineer booking. Nobody knows which step is slow.

(1) Pilot the chosen change in one region against the baseline.

(2) Map each step of the intake process as it actually runs.

(3) Ask Claude for options to speed up the slowest step, for managers to choose from.

(4) Measure the time spent at each step from ticket timestamps.

(5) Identify the bottleneck.

What is the correct order?

A 2, 4, 5, 3, 1

B 4, 2, 5, 3, 1

C 2, 5, 4, 3, 1

D 2, 4, 3, 5, 1

◆ PAUSE AND DECIDE

What is the correct order?

A 2, 4, 5, 3, 1

Map, measure, find the bottleneck, options, pilot.

B 4, 2, 5, 3, 1

Measures steps that have not been mapped.

C 2, 5, 4, 3, 1

Names a bottleneck before measuring anything.

D 2, 4, 3, 5, 1

Seeks options before knowing which step is slow.

THE SIGNAL

Map, measure, find, then fix, then pilot.

> The intake data is in. Triage takes 2.5 of the 5 days; engineer booking takes 1 day.
>
> The chief operating officer still proposes hiring six more engineers to fix the delay.

What is the best response?

A. Approve the hiring, because engineers are the most visible part of the service

B. Ask Claude to write a business case for more engineers

C. Focus on the triage step, with options piloted against the baseline

D. Let Claude run the whole intake process overnight with no one checking

◆ PAUSE AND DECIDE

What is the best response?

A Approve the hiring, because engineers are the most visible part of the service

Visibility is not where the time goes.

B Ask Claude to write a business case for more engineers

A persuasive document for the wrong fix.

C Focus on the triage step, with options piloted against the baseline

Targets the step that holds half the time, with evidence.

D Let Claude run the whole intake process overnight with no one checking

Removes human review from a client-facing process.

THE SIGNAL

The data names triage: fix the slowest step, not the loudest one.

> Halden piloted Claude-drafted triage in the North region for four weeks. Coordinators approved each draft. Report-to-booking time fell from 5.0 to 3.2 days.
>
> Nobody has checked whether priorities were set correctly.

What should happen before any wider roll-out?

A Roll out across all regions now, since the speed gain is clear

B Sample triaged tickets for wrong priorities against the baseline error rate, then decide

C Extend the pilot indefinitely to collect more speed data

D Ask Claude whether the pilot was a success

◆ PAUSE AND DECIDE

What should happen before any wider roll-out?

A Roll out across all regions now, since the speed gain is clear

Speed is half the story; quality is unmeasured.

B Sample triaged tickets for wrong priorities against the baseline error rate, then decide

Measures quality against the baseline, then decides.

C Extend the pilot indefinitely to collect more speed data

More of the same measure will not reveal the missing one.

D Ask Claude whether the pilot was a success

The evidence must come from the data, judged by people.

THE SIGNAL

A pilot that only measured speed has proved half: measure quality, then decide.

In five lines

- **Frame before you act:** the question, the objective, or the process map comes first.
- **Research:** frame, scope, gather, rate sources and confidence, verify, summarise.
- **Planning:** objective and phases before tasks; people commit owners and dates.
- **Process:** map, measure, find the bottleneck, options, pilot, decide.
- **Fix the slowest step, not the loudest.** Pilot against a baseline, measuring quality too.

COMING UP

Next: integrating Claude into a workflow.